LLM Engineer & Agentic Ai

RAG Systems • Agentic AI • FastAPI • LangChain

Lakki Marwat, KPK, Pakistan

PROFESSIONAL SUMMARY

Aspiring LLM Engineer with a solid foundation in Machine Learning (ML) and Deep Learning (DL). Currently specializing in Enterprise-grade RAG systems and Agentic AI workflows. Expert in building scalable AI applications using FastAPI, LangChain, and Vector Databases, with a focus on retrieval accuracy and latency optimization.

TECHNICAL SKILLS

Generative AI & RAG 2.0

- Architecture: RAG Full Pipeline (Chunking → Embedding → Retrieval → Generation)
- Advanced Retrieval: Hybrid Search (Dense + BM25), Neural Reranking (Cross-Encoders), Self-Querying
- Evaluation: RAGAS Framework – Faithfulness, Answer Relevancy, Context Precision
- Prompt Engineering: Zero-shot, Few-shot, Chain-of-Thought, and ReAct patterns

Core AI / ML

- Machine Learning, Deep Learning (PyTorch), and Convolutional Neural Networks (CNNs)
- Parameter-Efficient Fine-Tuning (LoRA / QLoRA) – Learning in Progress

Backend & Engineering

- Python Advanced: Async/Await, Decorators, and Generators
- API Development: FastAPI – Clean Architecture, Dependency Injection
- Data Validation: Pydantic v2 and Instructor Library for structured JSON outputs
- DevOps: Docker, Docker Compose, Git/GitHub (Branching & PRs)

Databases & Frameworks

- Vector Databases: ChromaDB, Qdrant, Pinecone
- Frameworks: LangChain, LangGraph, CrewAI

PROJECTS PORTFOLIO

Enterprise RAG Platform (Featured)

Built a system for querying private PDF data using Recursive Semantic Chunking and Cross-Encoder Reranking, achieving high faithfulness scores via RAGAS evaluation framework.

Multi-Functional AI Agent

Developed an autonomous agent using Tool Calling and Structured Outputs (Pydantic) to handle complex, multi-step workflows.

Computer Vision – Cat vs Dog Classifier

Implemented a Deep Learning model using PyTorch and CNN architecture, deployed as an interactive app via Streamlit.

ML Collection (5+ Projects)

A series of projects focusing on predictive modeling and data analysis using Python and Scikit-Learn.

KEY COMPETENCIES

- Optimizing LLM API costs and latency for production-ready applications
- Implementing LLM Caching (GPTCache) to reduce API bills by up to 40%
- Building stateful, cyclical multi-agent workflows using LangGraph
- Integrating diverse LLM providers: OpenAI, Anthropic, Gemini, and Groq

EDUCATION

12th Grade (Intermediate) – Completed

Self-Taught LLM Engineering Roadmap (2026) – Intensive specialization in Context Engineering, RAG 2.0, and Agentic Systems